

Compact AI HEALTH MONITORING And ANOMALY DETECTION DEVICE

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Early detection of health irregularities helps identify conditions before they develop, allowing timely intervention. Continuous monitoring, however, is not easy without wearable sensors that automatically track health data.

What doctors once did manually can now be handled by smart devices that show real-time vitals, detect subtle anomalies, and even predict possible health concerns. With advances in artificial intelligence (AI) and machine learning (ML), these systems can now learn normal health patterns and recognise even the slightest deviations as they happen.

In this device, the ML model runs on the edge, ensuring instant anomaly detection. The IndusBoard Coin manages processing, dataset collection, and ML model execution, with training performed through Edge Impulse.

The MAX30100 sensor measures SpO₂ and heart rate, while the compact 3cm IndusBoard Coin integrates an onboard battery connector and built-in sensors for temperature, magnetometer readings, and light. Its miniature form factor makes it well-suited for wearable health monitoring. Fig. 1 shows the author's prototype.

Circuit

The system is built around the IndusBoard Coin V2, interfaced with

a MAX30100 health sensor and an SSD1306 OLED display via the I²C bus, using default I²C pins (8 and 9).

EFY note. No vibration motor, buzzer, or LED has been included for anomaly alerts in this device, but

these can be added to any available pins and configured in the code.

Training ML model

To train the ML model, the first requirement is a dataset. In this device, the vitals of a healthy person

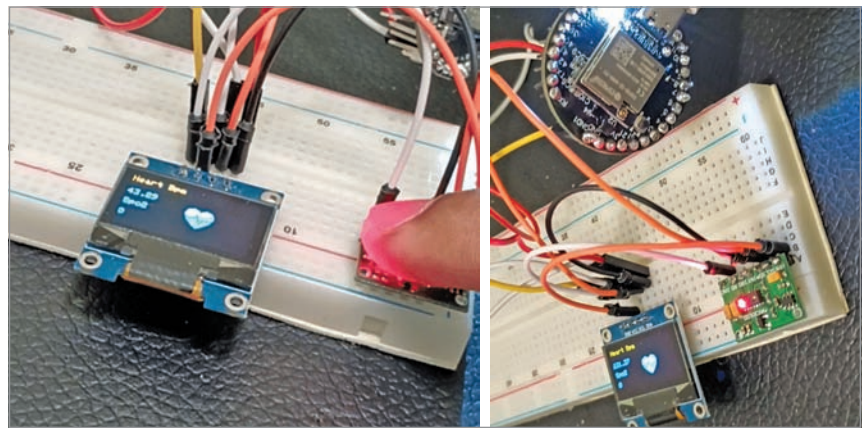


Fig. 1: Author's prototype

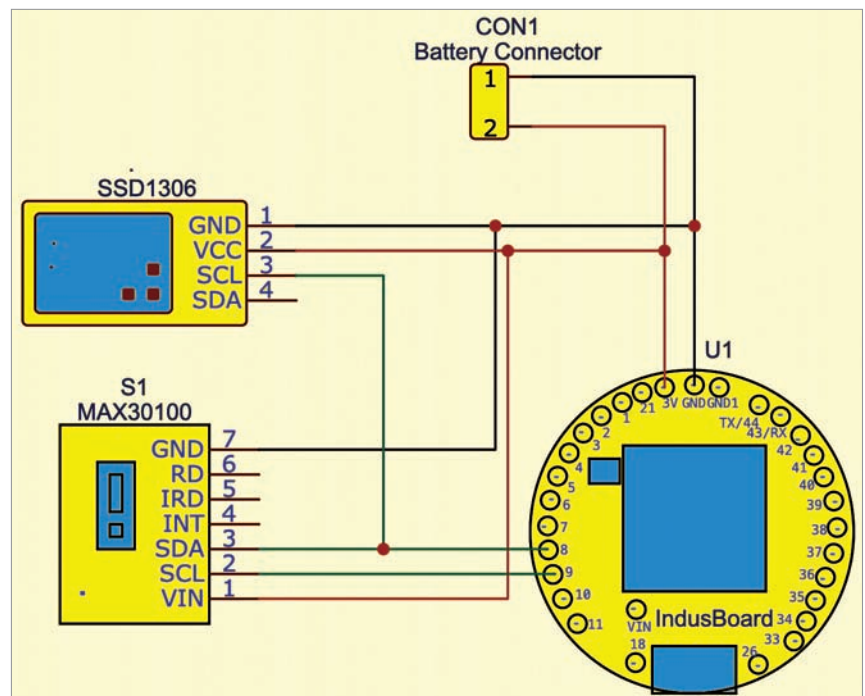


Fig. 2: Circuit diagram

BILL OF MATERIALS

Components	Quantity	Description
IndusBoard Coin V2	1	IoT board
Max 30100	1	Health sensor
SSD 1306 OLED	1	Display
3.3V battery	1	For power source